

Malhar Gudekar

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EDUCATION

University of Illinois Urbana Champaign

August 2024 - May 2026

Master of Science, Information Management

GPA: 3.8/4

• Coursework: Data Warehousing & Business Intelligence, Information Management, Information Consulting, Methods of Data Science

University of Mumbai

June 2019 - May 2023

Bachelor of Engineering, Computer Science

GPA: 9.25/10

• Coursework: DBMS, Big Data Analysis, Data Warehousing and Mining, Project Management, NLP, Machine Learning, Social Media Analytics

SKILLS

- **Analytical:** Data Analysis, Exploratory Data Analysis, Statistical Analysis, Trend Analysis, Root Cause Analysis, KPI Reporting, Data Validation, Data Cleaning, Anomaly Detection, ETL, Predictive Analytics
- **Programming / Querying:** Python, SQL, Pandas, NumPy, PySpark, PostgreSQL, Neo4j
- **Visualization / Reporting:** Power BI, Tableau, DAX, Excel, VBA, Dashboard Development, Data Visualization
- **Data / Cloud / Tools:** AWS S3, EC2, Redshift, EKS, FastAPI, Docker, Git, Kafka, pgVector, GCP

WORK EXPERIENCE

University of Illinois Urbana Champaign

May 2025 - Present

Research Assistant - iSchool

Champaign, USA

- Owned end-to-end design and development of cross-platform data integration system for 100+ users on AWS, enabling scalable data ingestion, extraction, and automated reporting
- Designed and optimized **PostgreSQL architecture** for high-volume, production-scale datasets, improving query performance by 38%
- Built robust, production-ready **Python** pipeline with retrieval and context summarization, improving multi-turn response accuracy by **43%**
- Developed **backend workflows** for real-time, fault-tolerant processing of longitudinal health data with focus on reliability and scalability

Research Assistant - CHI AI in Infodemic Management

January 2025 - May 2025

- Designed and deployed highly **scalable NLP pipelines in Python** to process high-volume datasets, reducing processing latency by **41%**
- Built distributed data ingestion pipelines using **Kafka** and **Neo4j** Streams for real-time, high-throughput processing and pattern detection
- Developed production-grade **APIs** and deployment workflows to improve system reliability and reproducibility
- Enabled large-scale data processing across social media streams using **distributed system architecture** principles

Business Intelligence Group

August 2025 - December 2025

Technical Consultant

Champaign, USA

- Built and deployed scalable, microservices-based Retrieval-Augmented Generation (**RAG**) system using **FastAPI**, **PostgreSQL**, and vector embeddings
- Designed efficient ETL and data migration pipelines to process, transform, and index large datasets, automating workflows to reduce manual reporting overhead
- Improved system performance by **64%** through optimized indexing strategies and pipeline efficiency improvements
- Developed backend services enabling real-time, high-performance querying of unstructured data

Swift Mobil Software Solutions Provider

July 2023 - October 2023

Data Analyst

Mumbai, IN

- Built interactive **Power BI** dashboards for **logistics** and **mobility KPIs**, reducing reporting time by **40%** and improving visibility into operational performance across transportation services
- Automated analysis of large-scale operational datasets using Python and PySpark to extract, identify trends, anomalies, and performance issues, combining root cause investigation with SQL-based reporting to support data-driven decision-making
- Developed **SQL** and **graph queries** to map dependencies, analyze operations, detect data quality issues, speeding root-cause analysis by **26%**

Pscope Technologies Pvt. Ltd.

January 2023 - June 2023

Data Analyst

Mumbai, IN

- Performed data validation, cleaning, and migration across large multi-source datasets using **SQL**, improving data accuracy by 30% and reducing reporting inconsistencies across sales, finance, and operations — directly supporting business and financial decision-making processes
- Automated recurring **Excel**-based reporting workflows using **VBA**, reducing report preparation time by **42%** and improving data consistency across weekly reporting cycles

Stu/Dio at Illinois

August 2025 - Present

Project Manager

Champaign, USA

- Led cross-functional team of **6** to deliver product updates for DeepCover, game, using user feedback and performance insights to guide data-driven feature decisions
- Tracked **sprint** progress, delivery metrics, and **QA issues** through **JIRA**-based workflows, improving cross-team visibility and supporting on-time milestone reviews

PROJECTS

FMCG Sales Demand Forecasting

March 2025

[View Project](#)

- Analyzed **FMCG sales demand data** across categories and locations to identify trends, seasonal patterns, and key drivers impacting inventory
- Built and compared forecasting models including **ARIMA**, **SARIMA**, and regression to evaluate demand patterns and improve forecast accuracy using RMSE and MAPE
- Derived actionable insights on promotions, pricing, and stock behavior, enabling better inventory planning and reducing stockouts and overstocking risks